Raj Srivastava

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Summary

Robotics student at NIT Rourkela (Integrated M.Sc., batch 2028). I build robots end to end, from PID balance control and sensor fusion to path planning and ROS2 arm control, and have won or placed at national competitions including IIT Bombay's e-Yantra and Gujarat Robofest.

Education

National Institute of Technology, Rourkela **Sep 2023 - Present**
Integrated M.Sc., Chemistry *Rourkela, Odisha*

Relevant Coursework

- Control Systems (PID, Feedback)
- ROS2 & Autonomous Navigation
- Embedded Systems, Embedded C
- Computer Vision
- Data Structures & Algorithms
- Operating Systems (Ubuntu)

Projects

Self-Balancing Builder Bot (e-Yantra, IIT Bombay, Runner-up) **Oct 2024 - Mar 2025**
C++, Arduino, PID Control, MPU6050 IMU, CoppeliaSim *GitHub* | *Video*

- Built a two-wheel self-balancing robot that works like an inverted pendulum, the same balance problem a humanoid solves on two legs. A real-time PID loop corrects tilt using live IMU readings.
- Added stable multi-directional motion, payload handling and servo-based pick and drop on the same chassis.
- Tested the control logic in CoppeliaSim before hardware and designed parts in Fusion 360. Placed Runner-up among 1200+ teams.

4-DOF Robotic Arm **Dec 2024 - Feb 2025**
C++, ROS2, MoveIt, Gazebo, Forward Kinematics *GitHub*

- Developed a 4-DOF arm with ROS2 joint-space control and forward kinematics, the same groundwork used for humanoid arm motion.
- Tuned PID gains for accurate pick and place and verified the full pipeline in Gazebo simulation.

Maze Solver Bot (Gujarat Robofest 5.0, 1st Place) **Oct 2025 - Mar 2026**
C++, ESP32, ToF Sensors, Computer Vision, Path Planning *GitHub* | *Video*

- Built a fully autonomous maze-solving robot on ESP32 that finished first at Gujarat Robofest 5.0.
- Fused a 4×4 multi-zone ToF array with side ToF sensors for live wall mapping, obstacle detection and node identification.
- Explored a 16×16 maze in about 50 seconds with Trémaux (DFS) exploration, then used Flood Fill (BFS) to cut a 34-cell route down to 7 cells.
- Replans the shortest path from the current node when a route is blocked, so the bot never has to restart.

Achievements

- **1st Place, Gujarat Robofest 5.0** **Mar 2026**
  Autonomous maze-solving robot with exploration, path memory and optimized navigation. Certificate
- **Runner-up, e-Yantra Robotics Competition (IIT Bombay)** **Mar 2025**
  Secured 2nd position among 1200+ teams. Certificate
- **Winner, e-Yantra Bot Race** **Mar 2025**
  First place in a timed, high-speed navigation challenge.
- **Top 2% Globally, LeetCode Biweekly Contest 186 (Rank 469 / 40,113)** **2026**
  200+ problems solved overall. Profile

Technical Skills

- **Languages**: C, C++, Python, Embedded C
- **Robotics & Frameworks**: ROS2, MoveIt, Gazebo, OpenCV, Sensor Fusion, Autonomous Navigation
- **Hardware & Embedded**: ESP32, Arduino, Raspberry Pi, NVIDIA Jetson Nano, PID Control, IMU and ToF Sensors
- **Tools & OS**: Linux (Ubuntu), Git, GitHub, VS Code, Fusion 360, CoppeliaSim

Extracurricular Activities

E-Cell, NIT Rourkela **Oct 2023 - Present**
Core Team Member *Rourkela, Odisha*

- Reached out to 50+ companies and secured sponsorships for the National Entrepreneurship Summit (NES), handling negotiations and formal communication.
- Led a team of 30+ members running startup and entrepreneurship events on campus.